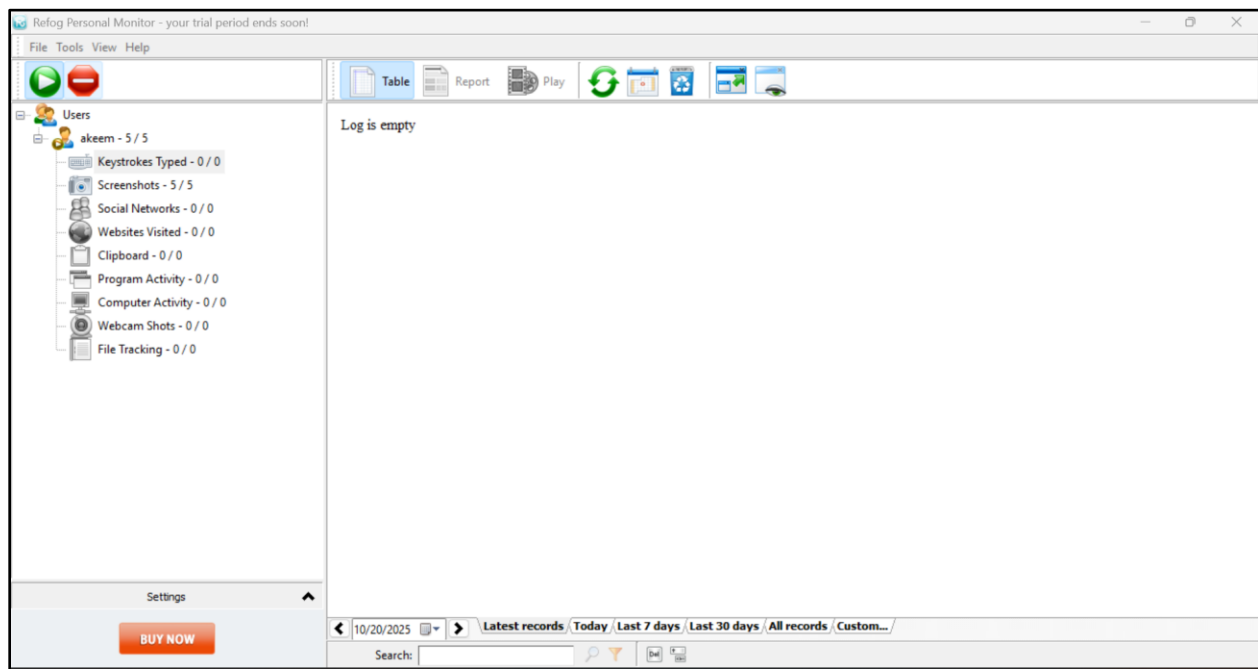


Refog, Spyrix, John the Ripper, or Prey Assignment

Question 1: what did I do?

Keylogging: Refog

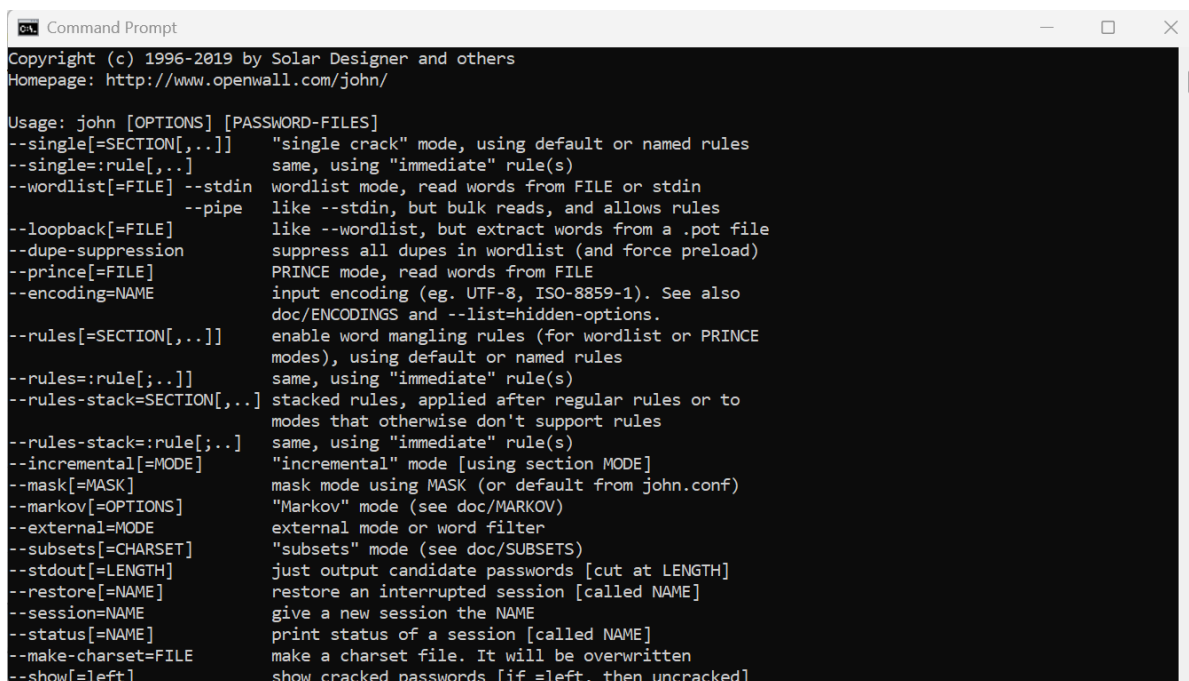
For the keylogging portion of this assignment, I selected **Refog** application to investigate. I had to temporarily disable my antivirus to complete the installation. I downloaded and installed the program, then enabled keystroke capture and screenshots from the left-hand menu. Refog is a commercial monitoring product marketed for parental and employee monitoring; however, it functions like any software keylogger and therefore poses substantive cybersecurity risks if misused. Refog captures user input by hooking into the operating system's input APIs, allowing it to record typed content, including passwords, and to take periodic screenshots that provide visual context for captured keystrokes. Depending on configuration, it also logs visited websites, launched applications, and files that are opened. The program runs in the background and can be configured to start automatically with the system; some versions offer an "invisible/stealth" mode so the software is not obvious to casual users. Refog (and similar keyloggers) may optionally upload logs to a remote dashboard so the person who installed it can review activity remotely.



John the Ripper

During my hands-on experiment, I installed **John the Ripper** after watching several instructional videos to understand the setup process. John the Ripper is a password-cracking tool designed to recover plaintext passwords from password hashes. It combines several attack methods, including dictionary attacks (using word lists of possible passwords), rule-based mutations, mask or pattern-based guessing, and brute-force attacks that attempt every possible combination. It optimizes the hash comparison process to test passwords rapidly against captured hashes. While the tool is used legitimately by cybersecurity professionals for password auditing and recovery, it can also be exploited by malicious actors. Importantly, tools like John the Ripper should only be used with explicit authorization to avoid ethical and legal violations.

John the Ripper operates by detecting the type of hash and input format within a file. Once a file containing password hashes is provided, it automatically identifies the hash type, groups them by salt (if applicable), and prepares optimized routines to attempt cracking. Some hashes, especially modern ones like bcrypt or scrypt, are intentionally designed to be slow to prevent rapid brute-force attempts. After detection, the program selects an attack mode, which can be dictionary, rule-based, mask, or brute force, and proceeds using the chosen approach.



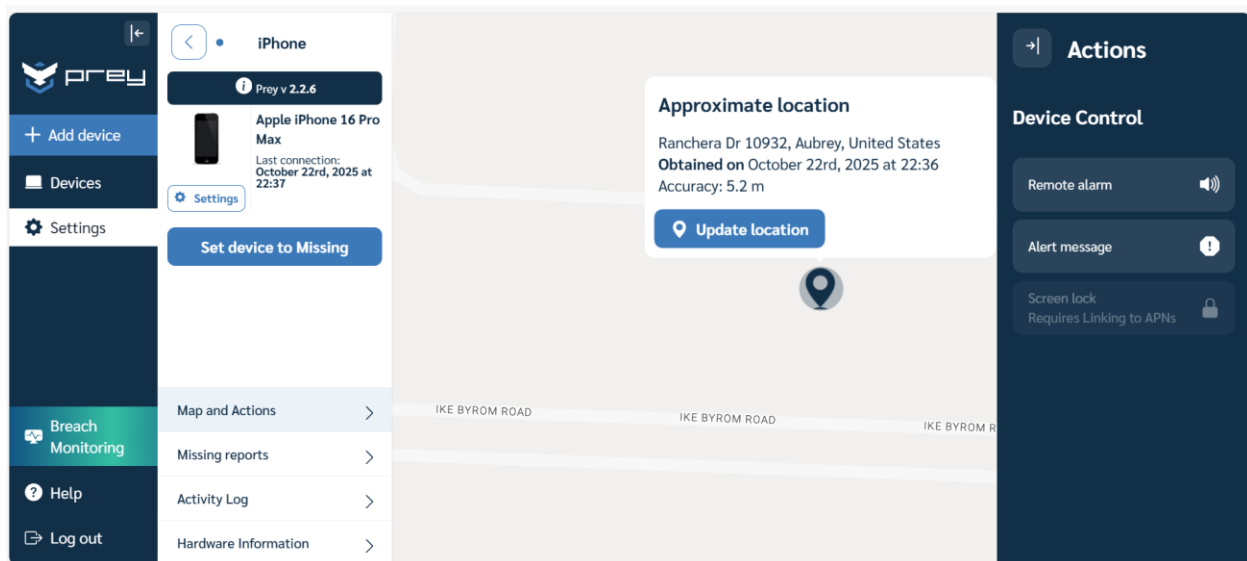
```
Copyright (c) 1996-2019 by Solar Designer and others
Homepage: http://www.openwall.com/john/

Usage: john [OPTIONS] [PASSWORD-FILES]
--single[=SECTION[,...]]  "single crack" mode, using default or named rules
--single=:rule[,...]      same, using "immediate" rule(s)
--wordlist[=FILE] --stdin wordlist mode, read words from FILE or stdin
                        --pipe like --stdin, but bulk reads, and allows rules
                        like --wordlist, but extract words from a .pot file
--loopback[=FILE]        suppress all dupes in wordlist (and force preload)
--prince[=FILE]          PRINCE mode, read words from FILE
--encoding=NAME          input encoding (eg. UTF-8, ISO-8859-1). See also
                        doc/ENCODINGS and --list=hidden-options.
--rules[=SECTION[,...]]  enable word mangling rules (for wordlist or PRINCE
                        modes), using default or named rules
--rules=:rule[,...]      same, using "immediate" rule(s)
--rules-stack=SECTION[,...] stacked rules, applied after regular rules or to
                        modes that otherwise don't support rules
--rules-stack=:rule[,...] same, using "immediate" rule(s)
--incremental[=MODE]     "incremental" mode [using section MODE]
--mask[=MASK]            mask mode using MASK (or default from john.conf)
--markov[=OPTIONS]       "Markov" mode (see doc/MARKOV)
--external=MODE          external mode or word filter
--subsets[=CHARSET]      "subsets" mode (see doc/SUBSETS)
--stdout[=LENGTH]        just output candidate passwords [cut at LENGTH]
--restore[=NAME]         restore an interrupted session [called NAME]
--session=NAME           give a new session the NAME
--status[=NAME]          print status of a session [called NAME]
--make-charset=FILE      make a charset file. It will be overwritten
--show[=left]            show cracked passwords [if =left, then uncracked]
```

Device Tracking with Prey

For the device tracking portion of this assignment, I used **Prey**, a client-and-cloud-based service designed to locate, monitor, and secure lost or stolen laptops, phones, and tablets. The system works by installing an agent on each protected device, which reports its location and system information to a control machine via the Prey web dashboard. From this console, users can activate “missing mode,” lock the device, wipe data, or view detailed location history. Prey determines location using a combination of GPS, Wi-Fi triangulation, and IP network data, which together generate accurate, real-time tracking information.

Once activated, the agent periodically reports the device’s GPS location, nearby Wi-Fi networks, IP address, and hardware details. If a device is marked as “missing,” Prey increases its reporting frequency and can capture screenshots or photos using the webcam to help identify its location or user. From the Prey web console, users can remotely lock the device, display a custom message, trigger an alarm, or even wipe sensitive data. Prey relies on GPS, Wi-Fi triangulation, and IP-based geolocation to track movement, and it stores a detailed location history for each device. However, it only works if the agent remains installed and the device is online.



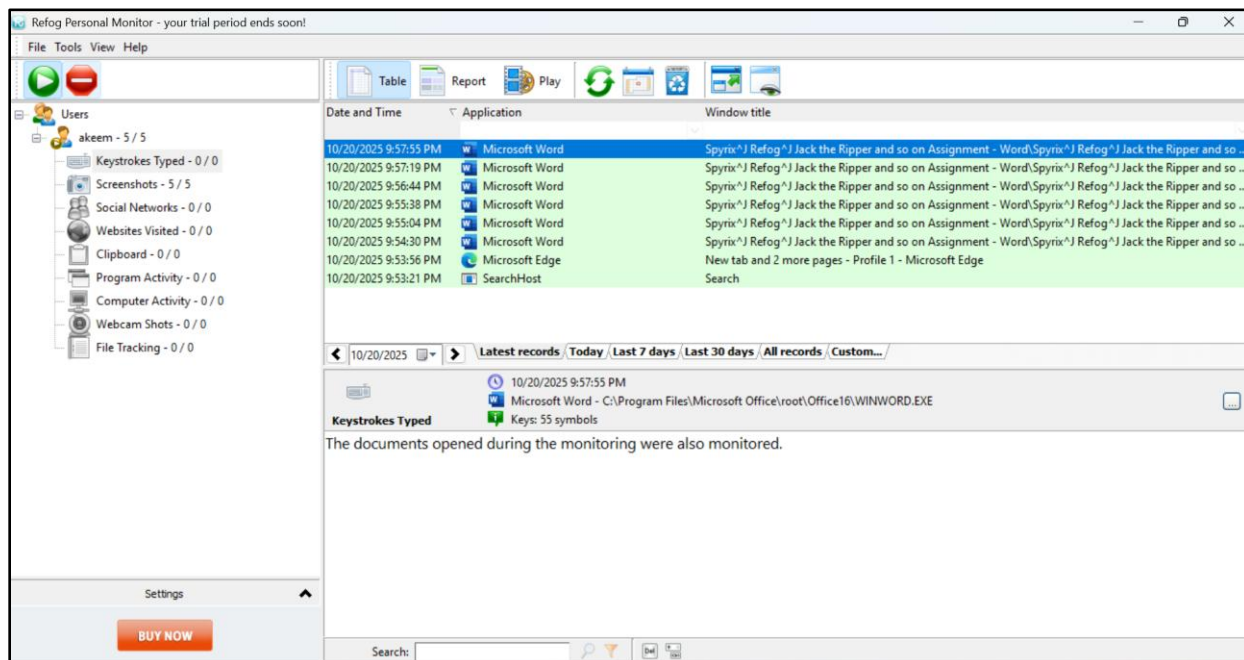
Question 2: what were the results?

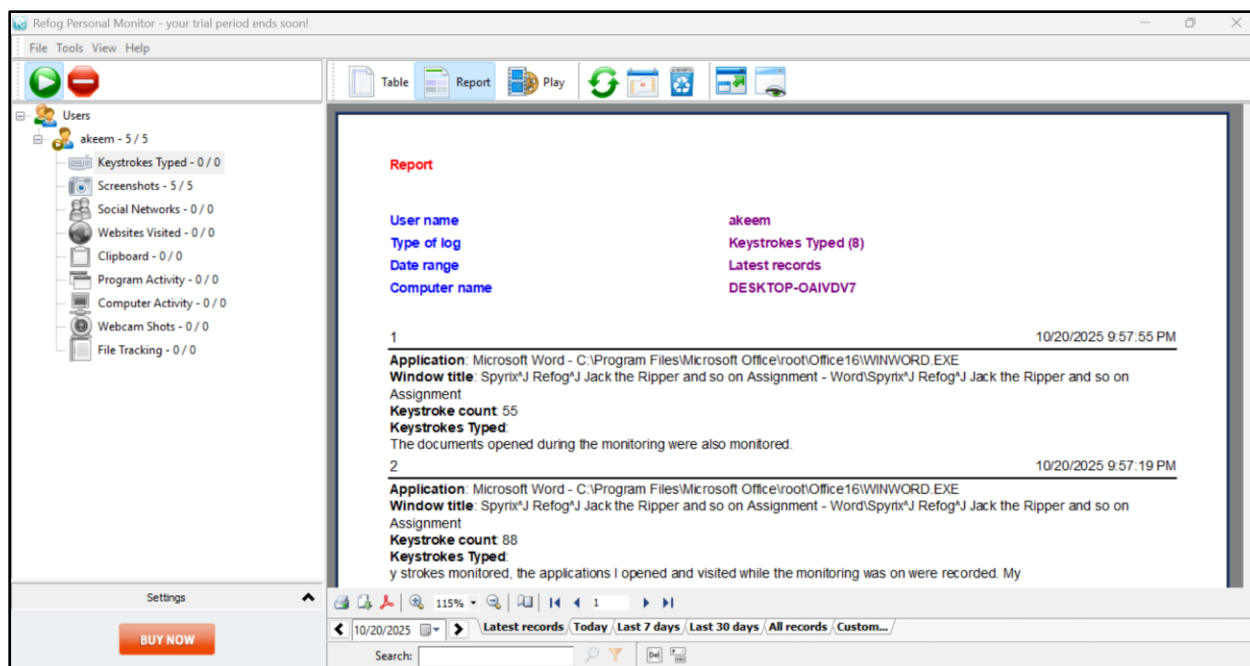
Refog

After selecting the keyboard-capture and screenshot options, I left Refog to monitor my activity for a few minutes. Within five minutes the software produced a detailed report showing the

number of keystrokes captured in each interval, the applications I opened, websites visited, and documents accessed while monitoring was active. The output demonstrated how comprehensive such monitoring can be even over a short period.

The ability to capture keystrokes, screenshots, and application or web activity means that keyloggers like Refog can be exploited by attackers to steal credentials, financial information, and other sensitive data. To minimize the risk of covert monitoring, users and organizations should maintain reputable and up-to-date antivirus and endpoint protection software, while also ensuring that operating systems are regularly patched. It is equally important to inspect running processes and startup programs for any unfamiliar or suspicious entries that could indicate hidden monitoring tools. Sensitive actions, such as changing passwords or performing online banking, should only be conducted on trusted and secure devices. Implementing multi-factor authentication, especially hardware tokens or app-based authenticators rather than knowledge-based questions, adds another layer of defense, making stolen credentials less useful to attackers. Additionally, using endpoint detection and response (EDR) solutions and centralized logging can help identify and remove stealthy monitoring software before it causes harm.





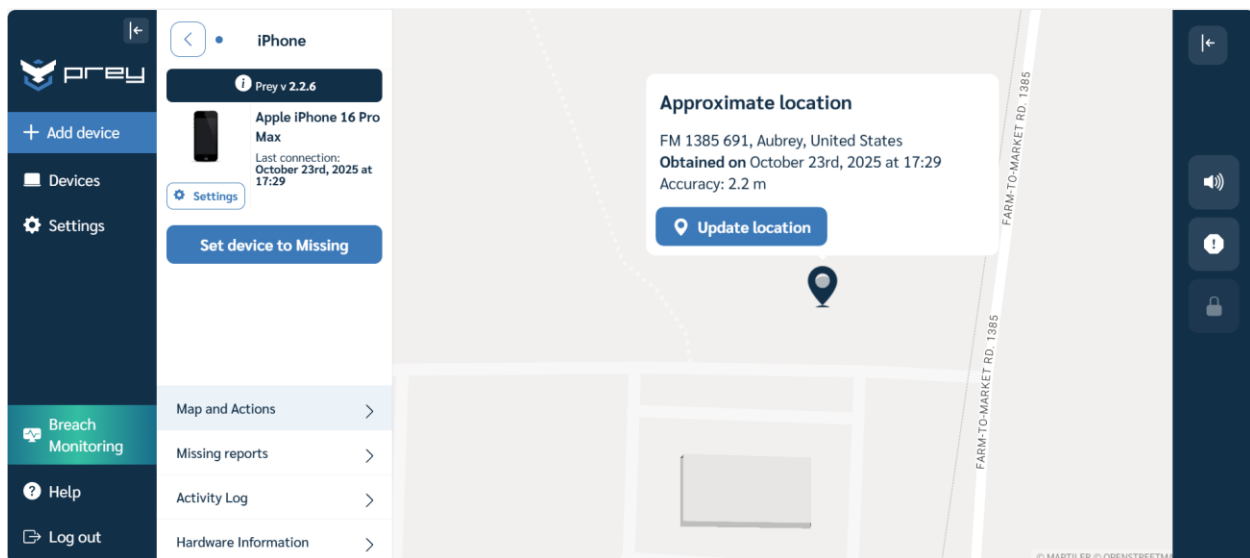
John The Ripper

After successfully installing **John the Ripper**, I decided to test its functionality by creating a password-protected PDF file and attempting to crack it. For my first trial, I used a complex password composed of eight alphabetic characters and ten numerals, making it both lengthy and difficult to guess. Once the brute-force attack began, the program continuously attempted different password combinations but ultimately failed to recover the original password. This outcome was expected, as the password's length and complexity significantly increased the computational effort required to break it. The test provided a clear illustration of how robust password design can effectively resist even advanced cracking tools.

For my second attempt, I simplified the password to see how the tool performed under less secure conditions. However, I encountered a technical issue, the program displayed an error message stating *"Crash recovery file is locked,"* which abruptly halted the process. This likely resulted from either improper configuration or my limited experience with the tool's recovery file management system. Despite the interruption, the experiment gave me a realistic understanding of how John the Ripper operates and the technical nuances involved in password-cracking tasks.

concerning such tracking tools can be: in the wrong hands, a malicious actor could use them to monitor someone's routines, movements, and physical locations.

The platform also supports remote actions, allowing users to lock the screen, display warning messages, trigger alarms, wipe files, or even initiate a full factory reset, depending on the operating system. Some configurations can even enable a "kill switch" that disables the device remotely.



Question 3: What did I learn?

Refog

From a cybersecurity perspective, the key takeaways from this exercise are quite significant. First, I learned how powerful and invasive keylogging software can be, even when marketed for legitimate purposes such as parental or employee monitoring. Tools like Refog demonstrate how easily sensitive data, such as passwords, banking details, or private communications, can be captured without the user's knowledge. This reinforced the importance of maintaining strong endpoint protection and regularly monitoring system activity for unusual processes.

I also learned that relying on antivirus software alone is not enough; good cybersecurity requires a layered defense that includes regular system updates, strong authentication methods, and safe user practices. The experiment highlighted why multi-factor authentication

(especially hardware or app-based methods) is essential in today's threat landscape, as it reduces the impact of stolen credentials. Finally, I realized how critical it is to stay vigilant, use trusted devices for sensitive actions, and develop an awareness of how everyday tools can become serious cybersecurity threats when misused.

John The Ripper

From a cybersecurity standpoint, this exercise was very insightful. It reinforced the critical importance of strong, complex passwords as the first line of defense against unauthorized access. Even with a powerful cracking tool, passwords that use a mix of uppercase and lowercase letters, numbers, and special characters, especially when combined into longer passphrases, are significantly harder to break. Implementing multi-factor authentication (MFA) adds another layer of protection, ensuring that even if a password is compromised, unauthorized access is still prevented. At the same time, the experiment highlighted how easily such tools could be misused by attackers who possess the right knowledge and computing resources. Watching several online demonstrations of John the Ripper being used to crack weak passwords effortlessly further emphasized how vulnerable simple or reused passwords can be.

Overall, this experience deepened my appreciation for the balance between usability and security in password management. It also reinforced the importance of adopting additional protective measures, such as multi-factor authentication, credential monitoring, and password managers, to complement strong password practices and safeguard against brute-force or dictionary-based attacks.

Device Tracking with Prey

This exercise revealed how powerful and intrusive modern tracking technologies can be. I learned that legitimate tools like Prey, which are designed for asset protection and recovery, can also pose serious privacy and security risks if misused. The ability to pinpoint exact locations, record movement history, and even access camera or screenshot data demonstrates how easily personal routines and sensitive information could be exposed. This emphasizes the need for strong access control, device encryption, and user awareness to prevent unauthorized installation or monitoring.

I also learned the importance of securing all devices with reputable antivirus software, keeping operating systems updated, and using multi-factor authentication to safeguard management dashboards like Prey's web console. Overall, this experiment reinforced that while location-tracking technologies can greatly enhance recovery and security, they must be carefully managed to protect privacy and prevent exploitation by malicious actors.